Pay to Quit

Robust Team Incentives with Hidden Need

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Software release

A company is shipping a major release built from many interdependent modules

Each engineer privately inspects her module:

절충안: 이것을 간단하게 줄이고

- Some modules have *latent bugs* 말로만 하지 말고도 스토리를 빨리 이야기 하는건? 의료 (수술), credible good, 안아프면 돈을 못 범 이것저것 찍어봐야 돈을 번다, 컨설팅?
- Others are *already stable*

소프트웨어가 놓치고 있는 것은?

incomplete information (adverse selection)

Team production, moral hazard 일반적으로 생각하는 조합 -> 0 -> 100을 만드는 것

If an engineer intervenes, she privately chooses her diligence:

우리는 예를 들면 30~50 -> 100을 만드는 문제를 생각하는 것 이게 안다뤄졌다

Thorough debugging (costly and effective) vs. a quick patch (cheap and ineffective)

What can the management do?

2페이지로 할 것

Pay only participants?

Common practice: Organizations reward workers who participate in projects

When workers differ in productive capacity

- Participation-based pay creates pooling
- Team size bloats beyond what is necessary, inflating externalities
- Principal must size bonuses for the worst case as if every module needs fixing

Separating contract?

- In organization, information is dispersed and experts may hold different beliefs
- How to separate robustly?

Question: How to optimally and robustly implement project success when uncertainty spans both types and actions?



This paper

Setting. Team production with complementarities, unobservable effort, and hidden types

Objective. Ensuring the maximum probability of project success,
robust to players' arbitrary beliefs about others' types and actions

Solution concept. Belief-free rationalizability (Bergemann & Morris, 2017)

Our findings. Optimal contract that achieves unique implementation of team project success

Today. Focus on fault-repair technology

Related literature

Contracting with externalities

- Segal (2003): divide-and-conquer under multilateral externalities
- Bernstein & Winter (2012): heterogeneous participation externalities

Team moral hazard with complementarities — unique implementation under action uncertainty

- Winter (2004): hierarchical contracts
- Halac, Lipnowski & Rappoport (2021): principal-designed uncertainty

Robust implementation — robust to beliefs about types

- Bergemann & Morris (2017); Dekel, Fudenberg & Morris (2007)

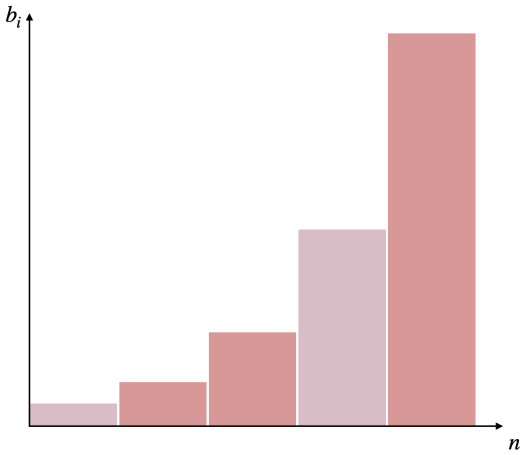
Mechanism design with evidence — exogenous evidence structure

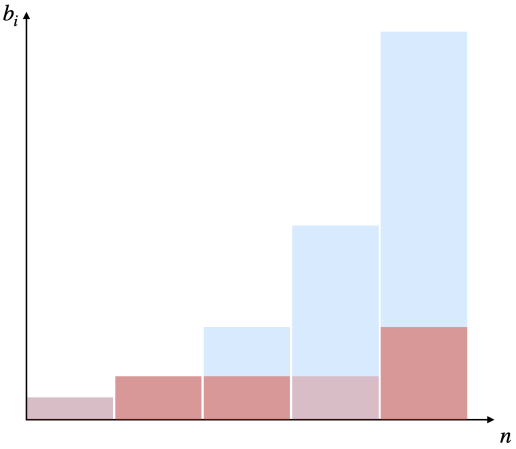
- Green & Laffont (1986); Bull & Watson (2007)

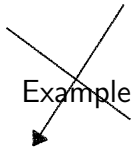
This paper: unique implementation robust to *both* type and action uncertainty, with *endogenous* evidence created through contract design.

Preview: When only 3/5 needs fixing

notebook LM 따라하기 유용한 동그라미







Example: Symmetric $n = 2$

A team of 2 experts

A project is managed by two experts, $i \in \{1, 2\}$

Types. Each privately observes $\theta_i \in \{L, H\}$ with $\Pr(\theta_i = H) = p \in (0, 1)$

Effort. Each then privately chooses $e_i \in \{W, S\}$, where W costs $c > 0$

Fault-Repair Technology.

Part quality

$$q(\theta_i, e_i) = \begin{cases} q_H & \text{if } \theta_i = H \\ q_H & \text{if } \theta_i = L \text{ and } e_i = W \\ q_L & \text{if } \theta_i = L \text{ and } e_i = S \end{cases}$$

Project success

$$P(\{1, 2\}) = 1, \qquad P(\{1\}) = P(\{2\}) = \tfrac{1}{2}, \qquad P(\varnothing) = \tfrac{1}{4}$$

Individual effort decisions are strategic complements

Principal's problem

The principal aims to implement all-high-quality ($q_1 = q_2 = q_H$) as a unique outcome and to do so at minimum cost

What can the principal condition on?

- Not effort — unobservable
- Not types — private
- But she can require a participation choice — observable, constrains subsequent actions

Participation as an observable margin

Each expert, after observing type, chooses whether to participate ($a_i = 1$) or opt out ($a_i = 0$)

- $a_i = 1$: Participants then privately choose $e_i \in \{W, S\}$
- $a_i = 0$: Opting out is effectively $e_i = S$

$\implies$ Actions: $(a_i, e_i) \in \{(0, S), (1, W), (1, S)\}$

We analyze a simultaneous-move game, where participation profile $\mathbf{a} = (a_1, a_2)$ is observed ex-post

Key features: opting out reveals shirking, so participation serves as partial evidence for effort

Contract. Bonus $b_i(\mathbf{a}) \geq 0$, conditioned on participation profile and project outcome

Payoffs and cost

Let $Q = \{j : q_j = q_H\}$

Expert i's payoff given a realization $(\boldsymbol{\theta}, \mathbf{a}, \mathbf{e})$:

$$u_i = P(Q)\, b_i(\mathbf{a}) \; - \; c\, \mathbf{1}\{e_i = W\}$$

Principal's expected cost under $\mathbf{p}$:

$$C_p(\mathbf{b}) = \mathbb{E}_{\mathbf{p}} \left[P(Q) \cdot \sum_{i \in N} b_i(\mathbf{a}) \right]$$

Solution concept?

Challenge: Complementarities $\Longrightarrow$ multiple equilibria

Types are private on top

Belief-Free Rationalizability (Bergemann & Morris, 2017) Def.

For each expert and for each of their types, iteratively delete actions that are never a best reply under any belief over others' surviving types and actions

Why BFR? Very permissive concept — weaker than BNE and ICR

Unique survival under BFR $\Longrightarrow$ unique under *every* belief model

Very robust!

Two contract forms

Two contract forms can be offered to each expert

- Participation-Only: $b_i(0, a_{-i}) = 0$. Non-participants are not paid
- Pay-to-Quit: $b_i(0, a_{-i}) > 0$. Non-participants are paid. $b_i(0, a_{-i}) > b_i(1, a_{-i})$ for non-trivial separation

Let's analyze two situations: (1) PO for both experts vs. (2) PtQ for both experts

Benchmark: Participation-Only

Participation-only contracts

Under PO, all types participate $\Longrightarrow$ Only team moral hazard remains (Winter, 2004)

Resolve experts in order (Divide-and-conquer)

Expert 1 with $\theta_1 = L$: W must dominate S, even at the worst case belief $Q_{-1} = \varnothing$

$$b_1 \cdot P(\{1\}) - c > b_1 \cdot P(\varnothing) \quad \Longleftrightarrow \quad b_1 > \frac{c}{P(\{1\}) - P(\varnothing)} \quad \Longrightarrow \quad b_1^* = 4c$$

Expert 2 with $\theta_2 = L$: Once rank 1's part is assured, 2's worst case belief $Q_{-2} = \{1\}$

$$b_2 \cdot P(\{1,2\}) - c > b_2 \cdot P(\{1\}) \quad \Longleftrightarrow \quad b_2 > \frac{c}{P(\{1,2\}) - P(\{1\})} \quad \Longrightarrow \quad b_2^* = 2c$$

Expected cost. $C^{PO} = P(\{1,2\}) \cdot (4+2)c = 6c$

Both experts paid regardless of type. Hierarchy sized for worst case $\boldsymbol{\theta} = (L, L)$.

Even if $\boldsymbol{\theta} = \mathbf{H}$, everyone participates and creates strategic uncertainty of size n.

Separation: PtQ contract

Pay to Quit: Round 0

Set $b_i(0, \mathbf{a}_{-i}) > b_i(1, \mathbf{a}_{-i})$ and start from $BFR_i^0 = \{(0, S), (1, S), (1, W)\}$.

Round 0. Type separation

For $\theta_i = H$:

- $(1, S)$ dominated by $(0, S)$: same quality, but $b_i(0, \mathbf{a}_{-i}) > b_i(1, \mathbf{a}_{-i})$
- $(1, W)$ dominated by $(1, S)$: same quality, saves c

For $\theta_i = L$:

- $(1, S)$ dominated by $(0, S)$: same quality q_L, but higher bonus

$$BFR_i^1(\theta_i = H) = \{(0, S), \cancel{(1, S)}, \cancel{(1, W)}\}, \qquad BFR_i^1(\theta_i = L) = \{(0, S), \cancel{(1, S)}, (1, W)\} \qquad \forall\, i$$

Participation assures quality

$$BFR_i^1(\theta_i = H) = \{(0, S)\}, \qquad BFR_i^1(\theta_i = L) = \{(0, S), (1, W)\} \qquad \forall\, i$$

Key observation: From Expert 1's perspective

- If $a_2 = 1$: only $(1, W)$ survives $\implies$ $q_2 = q_H$
- If $a_2 = 0$: type unknown $\implies$ q_2 can be q_H or q_L

$\implies$ Expert 2's participation assures $q_2 = q_H$ to Expert 1

PtQ: Round 1

Round 1. Expert 1 with $\theta_1 = L$ compares $(1, W)$ vs. $(0, S)$

$$b_1 \cdot \big[P(Q) - P(Q \setminus \{1\})\big] \geq c$$

Her IC depends on belief about Q, hence Expert 2's choice of a_2

If $a_2 = 1$**:** participation assures $q_2 = q_H$ in any belief $\qquad b_1^* = 2c$

If $a_2 = 0$**:** type unknown, worst case $q_2 = q_L$ $\qquad b_1^* = 4c$

$\Longrightarrow$ Expert 1's hierarchy shortens once her successor participates

(A small tie-breaking perturbation secures $(1, W)$ for the L-type)

$$BFR_1^2(\theta_1 = H) = \{(0, S)\}, \qquad BFR_1^2(\theta_1 = L) = \{\cancel{(0, S)}, (1, W)\}$$
$$BFR_2^2(\theta_2 = H) = \{(0, S)\}, \qquad BFR_2^2(\theta_2 = L) = \{(0, S), (1, W)\}$$

PtQ: Round 2

Round 2. Expert 2 with $\theta_2 = L$

Expert 1 already resolved $\implies$ $q_1 = q_H$ assured $\qquad b_2^* = 2c$

$$BFR_1^3(\theta_1 = H) = \{(0, S)\}, \qquad BFR_1^3(\theta_1 = L) = \{(1, W)\}$$
$$BFR_2^3(\theta_2 = H) = \{(0, S)\}, \qquad BFR_2^3(\theta_2 = L) = \{\cancel{(0, S)}, (1, W)\}$$

$\implies$ all L-types participate and work; all-high-quality uniquely survives

Expected cost.

$$C^{PtQ}(p) = P(\{1,2\}) \Big[\underbrace{(1-p)\cdot 2c + p \cdot 4c}_{\mathbb{E}[b_1^*]} + \underbrace{2c}_{b_2^*} \Big] = (4+2p)\,c < 6c = C^{PO}$$

Participation as endogenous evidence

Under PO, participation is **uninformative**: everyone participates regardless of type.

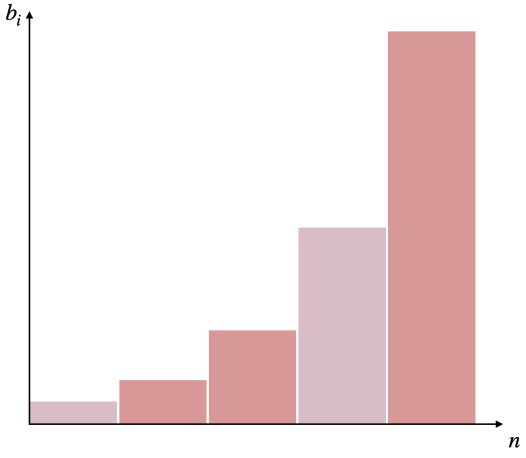
Under PtQ, the contract **endogenously creates** an evidence structure:

- Effort remains a hidden action the principal never observes
- Yet after Round 0, $a_i = 1$ becomes conclusive evidence for $e_i = W$ and $q_i = q_H$

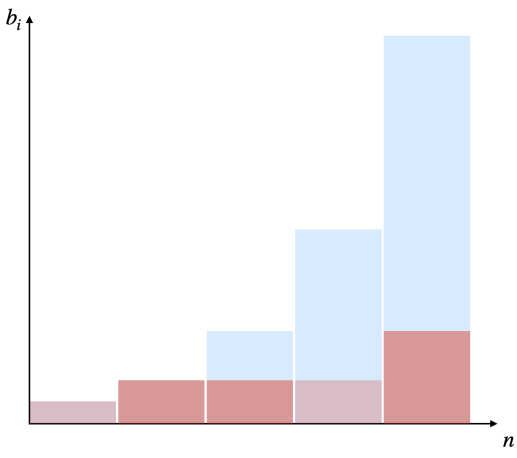
PtQ: Remarks

1. Each expert's strategic uncertainty = # later-ranked experts who opt out
2. **Expected cost** depends on the prior **p**
3. **Implementation** does not

PO vs. PtQ: When 3/5 needs fixing



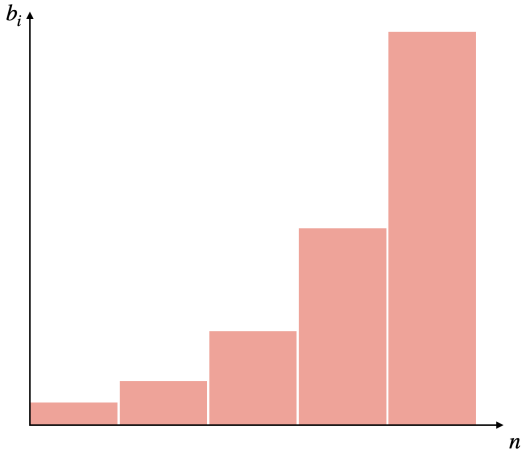
(a) PO



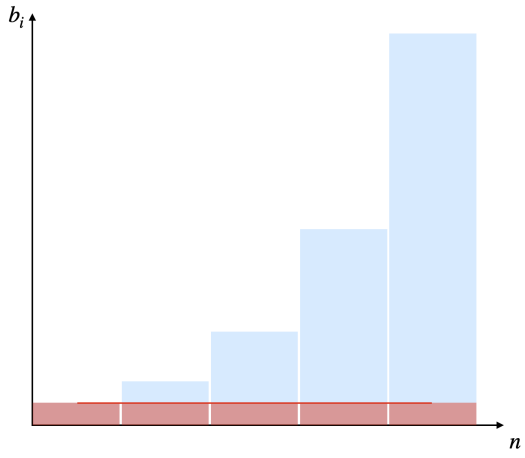
(b) PtQ

In PO, 5 participate and receive Winter payments with full length 5 hierarchy
In PtQ, 3 participate and the hierarchy reduces to 3 = (5 - 3 + 1)

PO vs. PtQ: When 5/5 needs fixing



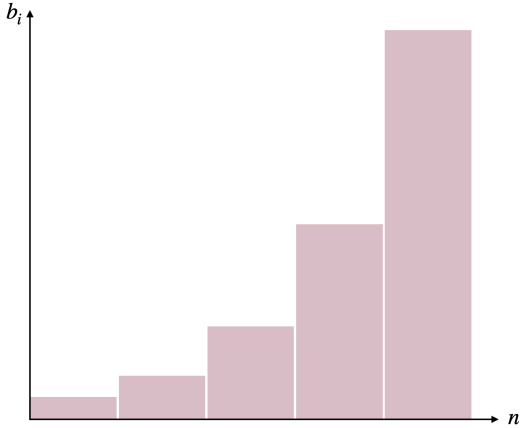
(a) PO



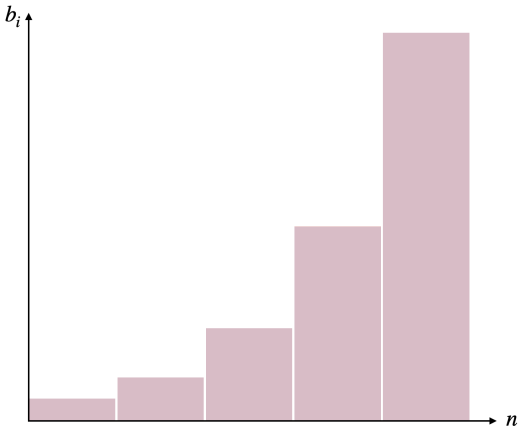
(b) PtQ

In PO, 5 participate and receive Winter payments with full length 5 hierarchy
In PtQ, 5 participate and the hierarchy reduces to 1 = (5 - 5 + 1)

PO vs. PtQ: When 0/5 needs fixing



(a) PO



(b) PtQ

In PO, 5 participate and receive Winter payments with full length 5 hierarchy
In PtQ, 5 opt out and the hierarchy is full 5

General Results

General setting

Environment

- n experts, heterogeneous type dist. $p_i = \Pr(\theta_i = H) \in (0,1)$ and effort costs $c_i > 0$
- Success probability $P(Q)$ is *strictly increasing* and *strictly supermodular* in the high-quality set Q:

$$P(Q \cup \{i\}) - P(Q) \;<\; P(Q' \cup \{i\}) - P(Q') \qquad \forall\, i,\ Q \subsetneq Q' \subseteq N \setminus \{i\}$$

Unique implementation (UI). **b** makes all-high-quality the *unique* BFR outcome

(up to a vanishing tie-breaking perturbation, since the UI set is typically open)

Optimal UI contract

$$\min_{\mathbf{b} \in \mathrm{UI}} \; P(N)\, \mathbf{E_p}\left[\sum_{i \in N} b_i\big(\mathbf{a}(\theta)\big)\right]$$

$n = 2$ easily extends

- **PO**: divide-and-conquer, a hierarchy of length n
- **PtQ**: quit bonus* = participation bonus*, with rank counting for later-ranked non-participants
  (1) every H opts out, then
  (2) each L is induced to participate and work, one by one along σ

Optimal contract?

- UI contracts need not be uniformly PO or PtQ across $\mathbf{a}_{-i}$
- But at each $(i, \mathbf{a}_{-i})$, $\mathbf{b}$ must either *pool* or *separate* i's type

  The design is a local choice between PO and PtQ

Optimal contract

Key spillover. At any $(i, \mathbf{a}_{-i})$, switching PO to PtQ

- is *free* for the principal, implemented at the *same* critical value
- is *strictly better* for the principal, as it will reduce the str. unc. of i's predecessors

Proposition (Optimal UI contract)

An optimal UI contract is a PtQ contract for all $(i, \mathbf{a}_{-i})$, for some $\sigma \in \mathrm{Perm}(N)$.

The problem collapses to finding an optimal ranking: $\sigma^* \in \operatorname*{argmin}_{\sigma \in \mathrm{Perm}(N)} C^{\mathrm{PtQ}}(\sigma, \mathbf{p})$

In general, σ^* admits no closed-form characterization.

▸ Proof

Fragility는 shirt vs opt out에서 하나의 경우 더 설명 이건 간단하게

그니까 우리가 고려하는 문제가 다뤄진적이 없는 모르는 문제라는걸 설명

이 위주로

Further results in the paper

1. O-ring production: a closed-form priority index characterizes the optimal PtQ ranking.
2. Common prior / BNE: PtQ can attain PI cost, but only under a specified belief structure.
3. Fragile technology: when careless intervention can damage a good part, pay-to-participate becomes an alternative separating contract.

Final remarks

- Paying experts to opt out is not to reward inactivity;
  but to create **endogenous evidence** that makes participation informative about quality.
- The informational content of an observable action therefore can be shaped by the incentive scheme.

Thank you!

Appendix

Special case: O-ring production

O-ring technology. $P(Q) = \prod_{i \notin Q} \lambda_i, \;\; 0 < \lambda_i < 1$
Every expert is a "generalist": i's marginal contribution $\propto P(Q_{-i})$

Closed form. Each later-resolved j inflates every predecessor's bonus by a loading $\kappa_j := (1 - p_j) + \dfrac{p_j}{\lambda_j} > 1$:

$$b^*_{\sigma(i)} = \underbrace{\frac{c_i}{1 - \lambda_i}}_{\text{base bonus}} \prod_{\sigma(j) > \sigma(i)} \kappa_j$$

Proposition (Optimal ranking under O-ring)

The optimal contract ranks experts by increasing priority index $I_i = \dfrac{c_i \, \lambda_i}{p_i (1 - \lambda_i)^2}$ *and offers PtQ for each* $(i, \mathbf{a}_{-i})$.

Rank early: *cheap* (low c_i), *critical* (low λ_i), *likely-already-good* (high p_i).
Symmetric λ: $I_i \propto c_i / p_i$. Symmetric λ & c: decreasing p_i (most-likely-good first).

▸ Proof

Fragile technology

Intervention without care *degrades* quality:

$$q(\theta_i, a_i, e_i) = \begin{cases} q_H & \text{if } \theta_i = H, a_i = 0 \\ q_H & \text{if } a_i = 1 \text{ and } e_i = W \\ q_L & \text{otherwise} \end{cases}$$

Now two contracts can separate types:

Contract 1 (PtQ): $b_i(0, \mathbf{a}_{-i}) > b_i(1, \mathbf{a}_{-i})$ (same as before)

- Participants = assured high quality. Better when defects are common

Contract 2 (PtP): $b_i(1, \mathbf{a}_{-i}) > b_i(0, \mathbf{a}_{-i}) > b_i(1, \mathbf{a}_{-i}) - c_i$

- Non-participants = assured high quality. Better when parts are mostly stable
- After type separation, hierarchical wage sizing for participants

Optimal: assign PtQ when $p_i \leq 1/2$, PtP when $p_i > 1/2$

Common prior and Bayesian NE

Under a common prior **p** and BNE:

PtQ under BNE achieves the partial-implementation cost

The principal can *tilt* bonuses: raise $b_i(1, a_{-i})$ when few participate, lower it when many do.
This preserves on-path expected payment while breaking all bad equilibria.

PO under BNE lowers cost relative to PO-BFR, but stays strictly above the PtQ-BNE cost.
Principal needs to compensate i for pooling j with $\theta_j = L$.

Cost ranking:

$$C^{PtQ,BNE} \ (= C^{PI,BFR}) \ < \ C^{PO,BNE} \ < \ C^{PO,BFR}$$

In either contract form, implementation is sensitive to belief structure.

PtQ: General characterization

$$b_i^* = \frac{c_i}{P(W_{-i} \cup \{i\} \cup R^\sigma_{<i}) - P(W_{-i} \cup R^\sigma_{<i})}$$

where $W(\mathbf{a}) = \{j \in N : a_j = 1\}$ are participants who are guaranteed to work, hence q_H and $R^\sigma_{<i}$ is the set of resolved experts before expert i.

Perturbations: Set $b_i(1, \mathbf{a}_{-i}) = b_i^* + \varepsilon_{i,1}$ and $b_i(0, \mathbf{a}_{-i}) = b_i^* + \varepsilon_{i,0}$.

$$1 < \frac{\varepsilon_{i,0}}{\varepsilon_{i,1}} < \frac{P(W_{-i} \cup \{i\} \cup R^\sigma_{<i})}{P(W_{-i} \cup R^\sigma_{<i})} \qquad \forall\, \mathbf{a}_{-i}$$

must hold to break ties to induce the desired action for each type

By strict monotonicity, $\frac{P(W_{-i} \cup \{i\} \cup R^\sigma_{<i})}{P(W_{-i} \cup R^\sigma_{<i})} > 1$ for all $\mathbf{a}_{-i}$. Such perturbations always exist

Belief-Free Rationalizability

Definition (Belief-free rationalizable actions)

Initialize: $B^{\mathbf{b}}_{i,0}(\theta_i) = \mathcal{A}_i = \{(0,S), (1,S), (1,W)\}$ for all $i \in N$, $\theta_i \in \Theta$

Iterate: $(a_i, e_i) \in B^{\mathbf{b}}_{i,k}(\theta_i)$ survives in $B^{\mathbf{b}}_{i,k+1}(\theta_i)$ iff there exists an arbitrary conjecture $\nu \in \Delta(\Theta_{-i} \times \mathcal{A}_{-i})$ such that:

(i) $\nu(\boldsymbol{\theta}_{-i}, \mathbf{a}_{-i}, \mathbf{e}_{-i}) > 0 \Rightarrow (a_j, e_j) \in B^{\mathbf{b}}_{j,k}(\theta_j)$ for all $j \neq i$

(ii) $(a_i, e_i) \in \arg\max_{(a_i', e_i') \in \mathcal{A}_i} \sum u_i^{\mathbf{b}}(\theta_i, a_i', e_i' \mid \boldsymbol{\theta}_{-i}, \mathbf{a}_{-i}, \mathbf{e}_{-i})\, \nu(\cdot)$

Limit: $B_i(\theta_i \mid \mathbf{b}) = \bigcap_{k \geq 0} B^{\mathbf{b}}_{i,k}(\theta_i)$

Back

Optimal UI contract: Proof sketch

Start from any UI contract $\mathbf{b}$:

1. Characterize: Let $\tau(i)$=elimination round for expert i and $S_i(\mathbf{a}_{-i})$=assured sets
2. Fill in: Choose a ranking σ refining (breaking ties in) τ
   Extend bonuses to all off-path profiles (unreachable) to critical values $\Rightarrow$ same cost
3. Reduce: Replace bonuses with critical values along σ $\Rightarrow$ weakly lower cost
4. Switch to PtQ: Expert by expert, PO-form $\rightarrow$ PtQ-form
   Each switch preserves UI and enlarges predecessors' assured sets $\Rightarrow$ weakly lower cost
5. Optimize: $C(\mathbf{b}) \geq C^{PtQ}(\sigma, \mathbf{p}) \geq \min_{\sigma'} C^{PtQ}(\sigma', \mathbf{p})$

Idea: Start from any UI contract.
We can always find a PtQ UI contract that is cheaper for each realization.

Back

O-Ring characterization proof

Under $P(Q) = \prod_{i \notin Q} \lambda_i$, expected cost under ranking σ:

$$C_{\mathbf{p}}^{P \text{ to } Q}(\sigma) = \sum_{k=1}^{n} \frac{c_{i_k}}{1 - \lambda_{i_k}} \prod_{\ell > k} \kappa_{i_\ell}, \qquad \kappa_i := (1 - p_i) + \frac{p_i}{\lambda_i}$$

Adjacent swap condition: rank i before j iff $\dfrac{c_i/(1-\lambda_i)}{\kappa_i - 1} \leq \dfrac{c_j/(1-\lambda_j)}{\kappa_j - 1}$

$\Longrightarrow$ Optimal ranking: increasing $I_i = \dfrac{c_i\, \lambda_i}{p_i (1-\lambda_i)^2}$.

Back